

# Gabriel Larot

Software Engineer

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## SUMMARY

Software engineer with experience building and evolving backend systems in production environments at scale. Focused on designing low-latency services, implementing decision logic, and maintaining reliable system behavior under real-world traffic. Comfortable working across service boundaries, handling edge cases, and improving systems through iterative development and feedback.

## EXPERIENCE

**Amazon** Seattle, WA  
*Software Development Engineer I* Mar 2025 - Present

- Built Java backend services for a large-scale Fire TV ads system, where filtering and selection logic operated under heavy production traffic across a distributed surface.
- Designed a real-time ad filtering flow for Fire TV Search using cross-account DynamoDB access and DAX caching to suppress low-relevance ads while preserving low-latency behavior.
- Expanded dynamic ad-selection workflows to additional sponsored placements, adapting logic for different eligibility rules, content availability, and fallback behavior while keeping outcomes consistent.
- Increased Dynamic Contextual Ads match success by about 2% by refining backend decision logic under strict latency constraints.
- Built a self-service ad relevancy analytics pipeline using Glue, Athena, QuickSight, and CDK, giving stakeholders historical visibility into scoring behavior across queries, placements, and positions.
- Strengthened deployment safety through composite-alarm monitoring, canary improvements, and rollout validation work that made backend changes safer under live traffic.
- Integrated product-discovery dependencies into backend hydration flows to support app-based sponsored content, including client setup, authentication changes, and service-to-service compatibility updates.
- Expanded validation coverage for backend service paths across marketplaces, reducing regression risk during launches and production rollouts.

**Data Annotation** San Jose, CA (remote)  
*Software Validator* Feb 2024 - Jun 2024

- Reviewed and evaluated AI-generated code across **Python, Java, SQL, and HTML**, focusing on correctness, maintainability, and performance tradeoffs in production-style scenarios.
- Identified logical errors and inefficiencies, improving overall code quality through structured review processes.

**Stanford University** Stanford, CA  
*Research Scientist, Intern* May 2022 - Sep 2022

- Modeled and simulated physical systems using **C++** and **MATLAB**, analyzing performance and stability across varying conditions.
- Wrote technical reports and presented results at internal meetings with Stanford's Plasma Dynamics Modeling Laboratory (PDML) doctoral candidates, identifying minor efficiencies in algorithmic designs.
- Evaluated tradeoffs in numerical models, improving runtime efficiency and correctness through iterative experimentation.
- Contributed software to support academic research workflows, translating experimental requirements into reliable, reusable code.

**U.S. Navy** Norfolk, VA  
*Aviation Ordnanceman* Aug 2016 - Aug 2021

- Operated in high-reliability, safety-critical environments requiring strict procedures, detailed troubleshooting, and accountability under pressure.
- Managed **\$500,000** worth of repair parts as the Repair Parts Petty Officer (RPPO) while maintaining **100% inventory accuracy** and ensuring timely procurement to support critical divisional operations.
- Diagnosed mechanical and system-level issues through structured analysis, documentation review, and coordination with cross-functional teams to restore operational readiness.

## EDUCATION

**San Jose State University** San Jose, CA  
*Bachelor of Science, Data Science* Aug 2021 - Dec 2024  
3.91 GPA

## SELECTED TECHNICAL WORK

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### Dynamic Ad Selection System (Amazon)

- Designed backend logic for real-time ad filtering and selection in Fire TV surfaces, with a focus on latency, correctness, and behavior under incomplete or degraded inputs.
- Extended dynamic contextual ad workflows into additional sponsored placements, reasoning through differences in eligibility rules, content availability, deduplication needs, and fallback paths.
- Improved live-system behavior by tightening decision logic and validating changes through staged rollouts, production monitoring, and structured debugging.

### Ad Relevancy Analytics Platform (Amazon)

- Built internal analytics workflows to surface historical ad relevancy behavior across queries, placements, and positions using Glue, Athena, QuickSight, Lambda, and CDK.
- Automated ingestion and parsing of S3-hosted data artifacts for downstream reporting, reducing manual analysis overhead for engineers and stakeholders.
- Improved operational visibility by consolidating dashboards and correcting metric references used to monitor production services.

### Deployment Reliability and Service Safety

- Improved deployment safety by adding composite alarms, bake-time monitoring, and rollback-aware validation paths for production changes.
- Increased canary effectiveness by refining severity-based alarm behavior and shortening validation intervals for faster issue detection.
- Added rollout controls that made deployment behavior easier to tune across environments while preserving production stability.
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## PROJECTS

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### Weather Impact on Bay Area Energy Consumption

[https://github.com/gabrield03/cs163/tree/main/src\\_sample/interactiveWebpage](https://github.com/gabrield03/cs163/tree/main/src_sample/interactiveWebpage)

- Designed and implemented a full-stack data application to analyze and forecast regional energy demand, integrating data ingestion, modeling, and visualization into an interactive system using **Python** and **Dash**.
- Built a data processing pipeline to aggregate and clean historical weather and energy datasets, aligning time-series inputs across multiple sources to support consistent model training and evaluation.
- Evaluated forecasting approaches using both statistical and deep learning methods, implementing **SARIMA** models for interpretability and **LSTM** networks to capture nonlinear temporal patterns.
- Compared model performance using error-based metrics, achieving **~7% error** relative to baseline consumption while analyzing tradeoffs between model complexity, interpretability, and generalization.
- Applied SHAP and Partial Dependence Plot (PDP) techniques to identify key weather features influencing predictions, improving transparency of model behavior and validating feature importance.
- Structured the application to use precomputed model outputs and optimized data loading, improving responsiveness and reducing runtime overhead in the web interface.

### Multithreaded Car Configuration Service

<https://github.com/gabrield03/Car-Configuration-Application>

- Designed a **thread-safe backend system** to manage shared configuration state under concurrent access.
- Implemented synchronization mechanisms (e.g., **locks/concurrent** data structures) to ensure safe state updates across threads.
- Identified and mitigated race conditions by enforcing atomic operations and controlled access patterns.
- Designed **REST APIs** for configuration updates and retrieval, ensuring consistent state across concurrent clients.
- Simulated concurrent workloads to validate system behavior under contention and evaluate tradeoffs between consistency and performance.

### Recipe Application

<https://github.com/gabrield03/RecipeApplication>

- Built a full-stack recipe-sharing application with backend APIs supporting reviews, voting, and user interactions.
- Designed relational data models and optimized query patterns, reducing query latency by **~20%** through indexing and efficient access strategies.
- Structured backend services to separate business logic and data access, improving maintainability and scalability.

### Sudoku Web Application

<https://github.com/gabrield03/Sudoku>

- Developed a backend-driven application with stateful sessions and validation logic enforcing Sudoku constraints.
- Designed server-side validation to prevent invalid moves and ensure consistent game state across user interactions.
- Managed persistent game state through API design, supporting reliable progression and repeat interactions.

## ENGINEERING APPROACH

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- Design backend systems with a focus on predictable behavior under live traffic, especially when inputs are incomplete or runtime conditions are messy.
- Favor incremental rollout strategies, validation coverage, and production monitoring when evolving distributed services.
- Approach backend changes through tradeoff analysis across correctness, latency, maintainability, and operational safety.
- Treat observability and deployment safety as part of the engineering work, not something added after implementation.

## TECHNICAL SKILLS

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**Languages:** Python, Java, SQL, TypeScript

**Frameworks / Libraries:** Spring Boot, Spring Context, JUnit, Mockito, TestNG, pytest, Lombok

**Cloud / Infrastructure:** AWS, AWS CDK, DynamoDB, DAX, Lambda, S3, CloudWatch, ECS, IAM, CloudFormation

**Data / Analytics:** Athena, Glue, QuickSight, Relational databases

**Backend & Systems:** Distributed systems, REST APIs, microservices, real-time decision systems, deployment automation, monitoring and observability, cross-account AWS architecture

**Tools:** Git, CI/CD pipelines, unit testing, integration testing, canary testing, change management workflows